Crimiviz – Milestone 1

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This is a preliminary milestone to set up goals for the final project and assess the feasibility of our ideas.

Dataset

- **Source & Content:** Our project utilizes the "Chicago Crime Dataset" (hosted on Kaggle, originally provided by the Chicago Police Department). It contains records of reported incidents, detailing the type of crime, location description, administrative boundaries (Districts, Wards, Community Areas), arrest outcomes, and precise geospatial coordinates (Latitude/Longitude).

  This dataset contains over 8.5 million rows of data from 2001 to the present. Each row represents one crime and includes details like the type of crime, the date, the location description, and whether an arrest was made.

- **Data Quality & Preprocessing:** The data is highly detailed, with a file size of 1.3GB. The dataset is frequently updated; as of writing this report, the most recent update was on February 23, 2026, meaning the data is current to within a month.

- **Efficiency:** We optimized the dataset by converting text columns (like crime types) into categorical variables, which significantly reduced memory usage.

- **Cleaning:** We noticed that only 1.11% of the data is missing spatial coordinates (latitude and longitude). We will remove these entries only when creating maps and evaluate whether these missing values follow the same distribution as the rest of the dataset or if they warrant separate analysis.

- **Formatting:** The dates need to be converted to a standard datetime format to accurately analyze trends by hour and month.

- **Feasibility:** Because 8.5 million points are too dense to display on a single map, we will employ techniques like "sampling" or "hexbinning" (grouping points into regional polygons) to ensure the visualizations remain smooth, responsive, and readable.

Problem Statement

General Topic & Axis: This project maps the temporal and spatial distribution of criminal activities in Chicago. The main axis of development is understanding the intersection of *when* crimes happen and *where* they happen.

What we are trying to show: We aim to reveal latent patterns that go beyond standard aggregate crime rates. Specifically, we want to visualize how different primary crime types (e.g., Theft vs. Deceptive Practice) aggregate in specific urban environments, and how the likelihood of an arrest fluctuates based on the neighborhood and the type of incident.

The core of the project will be an interactive map showcasing crimes with various selectors for the type of crime and hour of the day, alongside a dynamic visualization of the city's overall threat level across the day or year. Making our project visually appealing is a primary goal to simplify complex data, allowing users to intuitively grasp different spatial and temporal patterns.

Overview & Target Audience: By providing an interactive visualization of Chicago's crime hotspots and temporal trends, we aim to offer actionable insights. Our target audience comprises urban planners, public safety researchers, and city officials who need granular data to allocate municipal resources effectively and understand the shifting landscape of urban safety.

Exploratory Data Analysis

Our preliminary EDA, which can be found on our GitHub, provided several crucial insights for our upcoming visualizations:

- **Temporal Range and Volume:** To understand the "time horizon" of our dataset, this section identifies the earliest and latest recorded incidents and examines how the volume of crime reports has evolved annually from 2001 to the present.

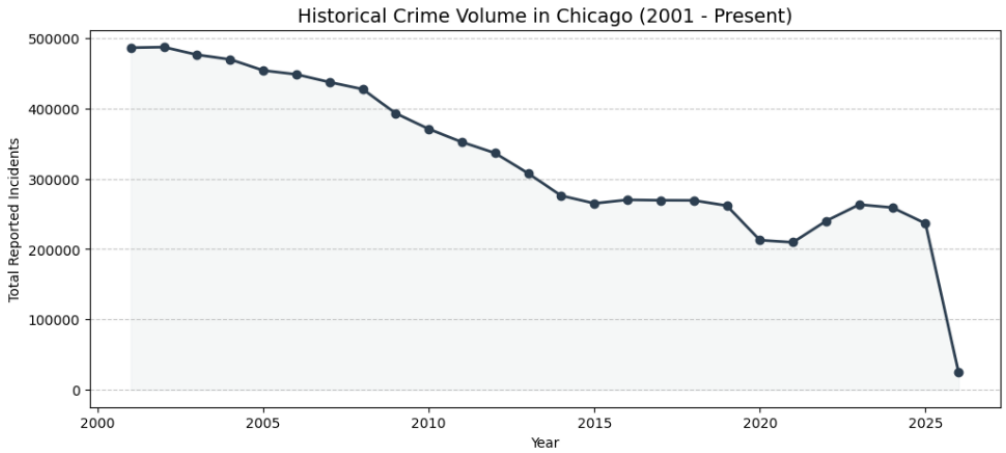

Figure 1: Evolution of the Number of Crimes Reported from 2001 to Present

- **Spatial granularity:** To optimize our map's "zoom level", we evaluated city divisions: Police Beats (305) risk visual clutter, while Districts (24) are too broad. Thus, Wards (50) or Community Areas (78) offer the optimal middle ground. Complementing this, the dataset includes 219 distinct Location Types (e.g., Residence, Sidewalk, Parking Lot), enabling highly specific spatial analysis across Chicago.

Table 1: Density of Crime Depending on the Type of Administrative Area

	Min Value	Max Value	Unique Count
District	1	31	24
Ward	1	50	50
Community Area	0	77	78
Police Beats	-	-	305

Table 2: Top Geographical and Location Types

Location	Number of Crimes
Street	2221334
Residence	1392463
Apartment	1017103
Sidewalk	765719
Other (215 Types)	269924

- **Categorization of Crimes:** Crimes are grouped into 34 primary categories, We also confirmed that the Primary Crime Category will be very relevant for our visualizations. However, the raw crime descriptions which are used only for deeper context might be too specific for our desired visuals.

  Primary Crime Categories: 34

  Detailed Crime Descriptions: 569

Examples of Descriptions: Forgery, Theft of lost or mislaid property, Illegal use of cash card

- **Crime Classification Analysis:** We discovered significant imbalances in crime types, some are rarely mentioned (e.g., Kidnapping, Stalking, Gambling), whereas others are vastly overrepresented (e.g., thefts account for more than 20% of the dataset).

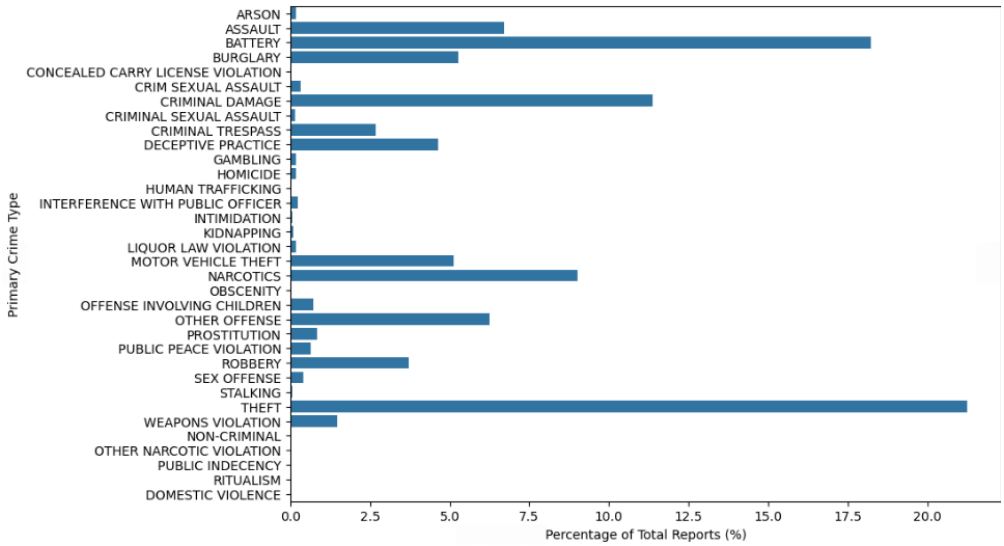

Figure 2: Distribution and Frequency of Primary Crime Classifications

- **Arrest Rate Disparities:** We calculated the global baseline arrest rate for the city (25.1%) and found striking variances when grouping by crime type. Compared to the average, certain crimes remain overwhelmingly "unsolved" at the time of reporting.

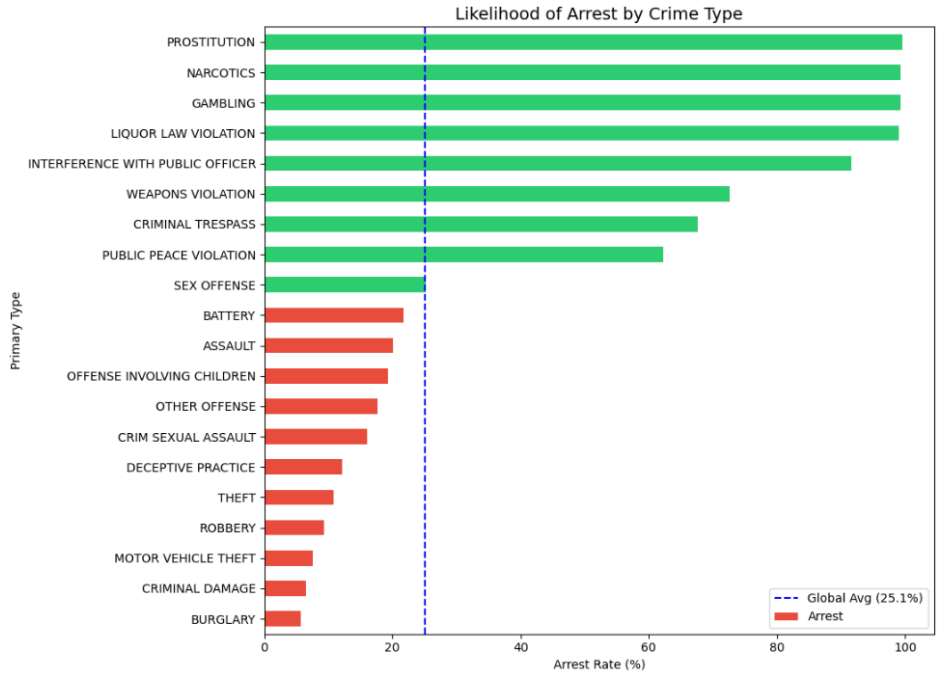

Figure 3: Disparities in Arrest Rates by Crime Type

- **Temporal Shifts (Insight 1):** Since Figure 1 indicated that the pandemic seemingly lowered the overall number of crimes, we structured a temporal comparison between pre-pandemic (2019) and post-pandemic (2023) data. This aims to identify structural shifts in crime locations (e.g., changes in Downtown vs. Residential incident volumes).

Table 3: Comparison of Pre-Pandemic and Post-Pandemic Crime levels

Year	2019	2023	Percentage Change
Street	56503.0	77679.0	37.477656
Residence	43524.0	31882.0	-26.748461
Apartment	35177.0	49455.0	50.589021
Sidewalk	20346.0	13052	-35.849798
Small Retail Store	7040.0	7266.0	3.210227
Restaurant	6984.0	5401.0	-22.666094
Alley	5119.0	6122.0	19.593671

Related Work

Chicago crime data is a classic public dataset frequently explored in basic data science tutorials, which typically stop at aggregate bar charts or simple static heatmaps. Our approach is original because we focus heavily on the interactive cross-filtering of spatial data with structural timeline shifts. A great illustration of existing work on this data can be found at `https://heyjackass.com/`. We want to expand on this type of analysis by showcasing the information on an interactive map. Instead of merely showing "where crime is highest," we aim to visualize the changing nature of crimes across types, time, and location.

- **Existing Work:** Many existing crime maps are visually cluttered because they display too many points at once, making them difficult to read or interpret effectively. They also often treat all times of day equally, which obscures important temporal patterns.

- **Originality:** Our approach is original because it prioritizes readability and user experience. We plan to explore deeper trends; for example, mapping whether specific locations in Chicago become more dangerous at specific times of the day. We can also explore peaks or troughs in crime linked to tourism or drops in incidents prior to specific city events.

- **Inspiration:** We draw inspiration from highly polished interactive maps, such as the NYT Census Map and The Pudding's 3D City Population Map. Regarding specific visualizations of our dataset, the most comprehensive work currently available is by the Chicago Police Department itself . However, their dashboard is highly technical, less visually attractive, and not particularly user-friendly. We want to pivot this perspective to make the data accessible and engaging for a broader audience.

- **Context:** This is our first time exploring this dataset. In the future, we may supplement or replace the static `chicago_crimes.csv` file with an API for real-time data integration.